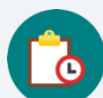


5.5 Drawing Conclusions From Given Information

Lesson Introduction

 Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg. MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.		 Learning Targets	- I can draw and mark figures using given information. - I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from about given information.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I use the information provided to mark figures appropriately? - How can I draw conclusions from given information?			
 Lesson Narrative	In this lesson, students will dive deeper into appropriately marking figures using given information. With the correct markings and other information, students will draw conclusions to solve problems.			
 Component Summary	5.5.1 Warm-Up (~5 min.)	Working through challenges (<i>SE p. 247</i>)	5.5.4 Collaboration (10 min.)	Collaboration on congruent overlapping triangles (<i>SE p. 251</i>)
	5.5.2 Collaboration (10 min.)	Collaboration on marking figures using given information (<i>SE p. 247</i>)	5.5.5 Try It (5 min.)	Practice drawing conclusions from given information (<i>SE p. 251</i>)
	5.5.3 Guided Instruction (10-15 min.)	Drawing conclusions from given information (<i>SE p. 249</i>)	5.5.6 Wrap-Up (~5 min.)	Ticket Out the Door on drawing conclusions from given information (<i>SE p. 252</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Colored Markers	
Additional Preparation - Work through each of the problems in advance to follow the flow of the lesson to help with facilitating the collaborative activities. 5.5.1 Warm-Up, contains a video to accompany the question prompts				

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may misread the labeling and make incorrect markings. - Students may not make their markings unique, leading to confusing information. - Students may not struggle with overlapping triangles.	- In this lesson, students are drawing information from given shapes. Students may struggle with reading and interpreting the information correctly. Consider using colored highlighters or markers to help link each identified piece of information that can be connected between the image and the statements. - If time is an issue, consider assigning 5.5.5 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 5.5.2 and 5.5.4 Collaborations provide opportunities for students to work together. Students will be able to try the problems on their own and then pair up with their partner to share their work with one another and come to a consensus.	MA.K12.MTR.5.1: Patterns and Structure Students will be drawing conclusions from given information throughout this lesson. Students will need to make appropriate markings and complete various components of the lesson to solve more complex problems.
 Differentiate Instruction	Consider pairing students based on their performance in the prior set of lessons. Student’s understanding of what they have learned so far about triangle congruence can be a way to determine which students may need additional support. Consider the use of physical manipulatives during 5.5.4 Collaboration for visual learners to be able to visualize the various components when working with overlapping triangles. 5.5.6 Wrap-Up includes a Ticket Out the Door that provides students the opportunity to display mastery of the content from this lesson. Student response data can provide information on the degree to which students have mastered the content of this lesson. Consider evaluating each part of the question separately when collecting data to evaluate understanding.	
Lesson Notes:		

5.5.1 Warm-Up: Working Through Challenges

Component Overview: Students engage in an activity about working through challenges. For this activity, show the video for a more in-depth conversation during the interview with the student, Teri.



5.5.1 Warm-Up: Working Through Challenges

Teri takes dance in school, but learning how to tap dance was a new and overwhelming experience. She said that after her teacher introduced tap, “It was really hard for me to learn ... It just looked like a lot of work and looked like it took a lot of time to learn, and it was really high tempo.”

1. What is something that felt overwhelming the first time you were learning it?

Answers will vary. I found learning a foreign language to be overwhelming the first time I tried to learn it.

Teri’s teacher slowed down the moves to show her the structure and recommended that she practice each step before trying to put it all together. She connected this experience to learning other things that are difficult, saying, “The more you practice a certain skill that you need, the better you become at it, and the easier it becomes.”

2. Describe a time when practicing something that was challenging for you helped you become better at it.

Answers will vary. I practiced a musical instrument and over time was able to get much better at it. I practiced a sport and over time was able to get much better at it. I practiced a video game and over time was able to get much better at it.



For this activity, consider modifying this to be a think-pair-share activity. Since this does not require any specific mathematical knowledge, students who may not often participate will have the same entry point as others and be able to engage in the conversation.



Consider having students pair up with a partner to share their answers to questions 1 and 2. Allow each partner to share their partner’s story to improve listening and communication skills in conjunction with the discussion on working through challenges.

5.5.2 Collaboration: Using Given Information to Mark Figures

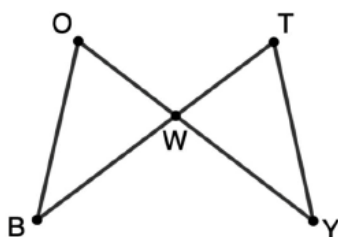
Component Overview: Students will work together to determine how to accurately mark figures using given information. For this activity, have students work with a partner to complete question 1 and then review answers together as a class. Then, release students to complete questions 2 and 3.



5.5.2 Collaboration: Using Given Information to Mark Figures

When given information about geometric figures, it is helpful to mark the figures.

1. Triangles OBW and WTY are shown.



Josue and Abby marked the figure in different ways given $\overline{OW} \cong \overline{TW}$, $\overline{OB} \cong \overline{TY}$, and $\angle OBW \cong \angle TYW$ as shown.

Josue's Markings	Abby's Markings

- Discuss with your partner what you notice about how Josue and Abby each marked the figure.
- Complete the table to describe similarities and differences in the way each student marked the figure.

How were their markings similar?	How were their markings different?
Both methods clearly show congruent angles and sides.	Josue used different colors to identify congruent sides and angles. Abby used tick marks to identify congruent sides and angles.



Notice and Wonder

- What are some things that you notice about the triangles shown in question 1?
- What are some things that you wonder about the triangles shown in question 1?



There are many ways that students can mark figures. Two examples are shown (using colors and using tick marks). There is not one “right” way. Rather, it is important that students have a strategy they understand and use to mark figures after reading given information.



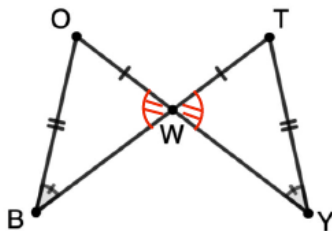
It's helpful to have highlighters on hand if students prefer to use color to mark corresponding parts.



When reviewing answers, have students share their responses to both the similarities and the differences. Consider displaying answers that are being shared. Help students to realize that the method used for marking is not as important as the accuracy of the markings themselves. Allow students to know that they can choose a method that best works for them.

5.5.2 Collaboration: Using Given Information to Mark Figures (continued)

- c. Josue wondered if there was enough information given to conclude $\triangle OWB \cong \triangle TWY$. Work with your partner to explain whether there is enough information to conclude $\triangle OWB \cong \triangle TWY$.
Answers will vary. No there is not enough information. $\angle WOB$ can differ from $\angle WTY$, and this would change the length of sides $\overline{WB}$ and $\overline{WY}$ respectively. Therefore, SSA is not able to prove triangles congruent.
- d. Abby said she does not think there is enough given information to conclude $\triangle OWB \cong \triangle TWY$, but she said there is enough information to conclude $\angle OWB \cong \angle TWY$. What definition is Abby using to draw this conclusion?
Abby used the Vertical Angles Theorem.
- e. Add markings to Abby's figure to represent her conclusion that $\angle OWB \cong \angle TWY$.

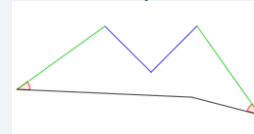


- f. Work with your partner to explain whether there is now enough information to conclude $\triangle OWB \cong \triangle TWY$.
Answers will vary. Yes, now that I know two angles and at least one side congruent, there is enough information to conclude congruence with Angle-Side.



On question 1c, to provide a visual understanding of why not enough information was given, draw the figure with the congruent parts and with the other parts purposefully not congruent.

For example:



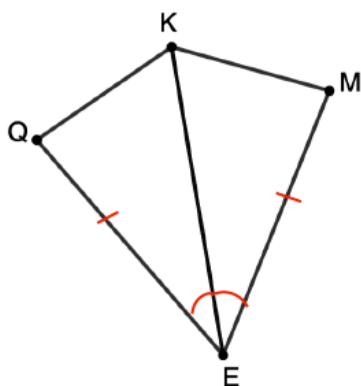
- How did you determine which markings could be added to the figure?
- What could be used to justify that your markings are correct?



Consider splitting the class into four groups, each with a different congruency postulate or theorem. Then have each group either prove why the triangles could be proven congruent or why they could not be proven congruent, using their assigned postulate or theorem.

5.5.2 Collaboration: Using Given Information to Mark Figures (continued)

2. Mark polygon $KQEM$ using the given information.
 $\overline{EQ} \cong \overline{EM}$ and $\angle KEQ \cong \angle KEM$



Students could use one of the strategies modeled from Josue's or Abby's work, or they could use another system of marking that works for them. Keep an eye out for ways that students mark their figures and have students share their strategies if any different ones are used.

3. Compare your markings with your partner. Make any corrections if necessary.

5.5.3 Guided Instruction: Drawing Conclusions From Given Information

Component Overview: Students will receive guided instruction on drawing conclusions from given information. For this activity, guide students through each question in the activity.



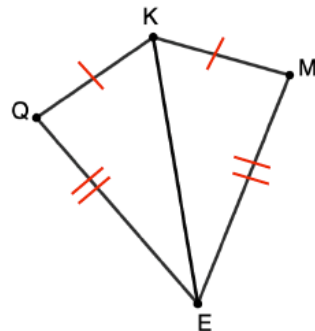
5.5.3 Guided Instruction: Drawing Conclusions from Given Information

Given information, definitions, properties, postulates, and theorems can be used to draw conclusions about figures.

1. What are some definitions, properties, postulates, and theorems related to triangles and angles learned so far this year?

Definitions	<i>Answers will vary. midpoint, perpendicular lines, perpendicular bisector, right triangles, angle bisector</i>
Properties	<i>Answers will vary. Reflexive property of congruence, transitive property, all angles of a triangle add up to 180°.</i>
Theorems and Postulates	<i>Answers will vary. SSS, SAS, ASA, AAS, HL, Vertical Angles Theorem, Corresponding Angles Theorem</i>

Polygon $KQEM$, where $\overline{KQ} \cong \overline{KM}$ and $\overline{EQ} \cong \overline{EM}$ is shown. $\overline{KE}$ is drawn inside polygon $KQEM$.



2. Mark the figure using the given information.
3. Discuss any conclusions that can be drawn about the figure with the information given.



Question 1 is not meant to be an exhaustive list; rather, an opportunity to brainstorm concepts that may come into play when drawing conclusions. This should also not take a lot of time but provide a quick primer for the remainder of the lesson. Students can add to this table as additional definitions, properties, postulates, and theorems are used throughout the lesson.



Consider using question 1 as a whole class share-out activity. For example, an entry point that is accessible to all is a silent board activity. Put the three categories on the whiteboard and have students come up silently to add an item not already listed as they think of each new item. Students can choose which category to write in. You can also create a game out of this by creating two teams and having students add items one at a time until the team can no longer add anymore, giving the team a point for being the last one to add a definition, property, or theorem and postulate.



Give a little extra time for students to share their markings and conclusions in questions 2 and 3 with all the students in the class. Getting students to understand these two elements is critical to learning the concepts of this unit.

5.5.3 Guided Instruction: Drawing Conclusions From Given Information (continued)

Using the given information, Hilary concludes $\triangle QEK \cong \triangle MEK$.

4. Complete the statement to explain how Hilary is able to draw this conclusion.

Hilary noticed ☐ 1
☒ 2
☐ 3 corresponding ☐ side(s)
☐ angle(s)

of $\triangle QEK$ and $\triangle MEK$ are congruent based on the given information.

Both triangles also share a common ☐ side,
☐ angle, which is

congruent to itself because of the

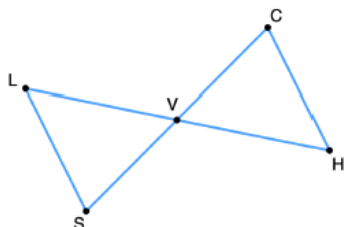
☐ reflexive
☐ transitive property.

She can conclude $\triangle QEK \cong \triangle MEK$ because of

☐ SSS.
☐ SAS.
☐ ASA.
☐ AAS.
☐ HL.

5. Triangle LVS and triangle CVH are shown.

$$\overline{SV} \cong \overline{VC} \text{ and } \angle LSV \cong \angle HCV$$



- a. What conclusions can be drawn from the given information?

Answers will vary.

Triangles VLS and VHC are congruent since $\angle LVS$ and $\angle HVC$ are vertical angles and vertical angles are congruent by the Vertical Angles Theorem.



To set students up for success with what they'll be asked to do, consider asking the following questions:

- What conclusion(s) did Hilary draw?
 - $KE = KE$
 - the triangles are congruent
- Why was she able to draw those conclusions?
 - reflexive property
 - SSS



Allow students to share the conclusions they developed in question 5a. Consider posting various conclusions on a piece of chart paper or on a section of the whiteboard. Allow students to debunk any conclusions based on evidence.



What knowledge was necessary to make additional appropriate markings?

5.5.3 Guided Instruction: Drawing Conclusions From Given Information (continued)

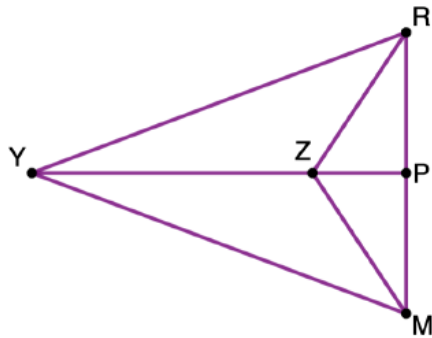
- b. What congruence postulate or theorem is the reason these conclusions can be made?

Angle-Side-Angle

- c. Julia says she can conclude $\overline{CH} \cong \overline{SL}$ and $\angle SLV \cong \angle CHV$. Explain why Julia is able to draw these conclusions.

Answers will vary. These conclusions can be drawn because once triangles are proven congruent, all corresponding parts will be congruent.

6. Triangle RMY is shown. $\angle YMZ \cong \angle YRZ$ and $\angle RZY \cong \angle MZY$.



Work with your partner to complete the statements based on the given information.

- a. I can conclude ...

$\triangle YRZ \cong \triangle YMZ$

- b. I know this because ...

By the reflexive property of congruence $\overline{ZY} \cong \overline{ZY}$ so $\triangle YRZ \cong \triangle YMZ$ by AAS.



Notice and Wonder

- What do you notice about the triangle in question 6?
- What do you wonder about it?
- How would you go about marking this triangle?



Have students compare their answers in question 6a and 6b with a partner. Then have each partner pair compare their work with another partner pair.

5.5.4 Collaboration: Overlapping Triangles

Component Overview: Students will work together on problems that involve overlapping triangles. For this activity, allow students to work on both questions with a partner.

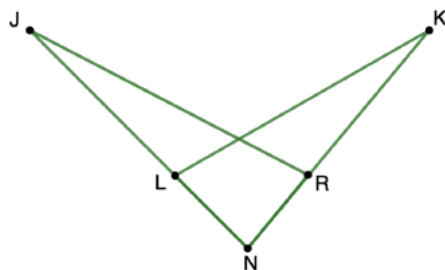


5.5.4 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

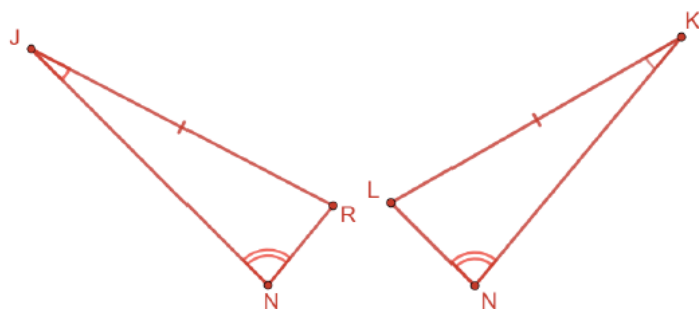
Dalton saw the following question with the polygon shown in an old geometry textbook.

"If $\angle NKL \cong \angle NJR$ and $\overline{KL} \cong \overline{JR}$, what else do you know so that you can show that $\triangle RJN \cong \triangle LKN$ using ASA?"



He knew that both triangles share $\angle N$. Therefore, by knowing that $\angle N \cong \angle N$ by the reflexive property of congruence, he concluded that he could use ASA to show that $\triangle RJN \cong \triangle LKN$.

1. Draw the triangles so that they do not overlap. Then, mark the triangles with the information given from the textbook.



When reviewing the answers for this activity, begin by reviewing the context and image provided at the beginning of this activity. Provide students with a think aloud about things you notice about the information.



Consider using patty paper to provide a visual of how to properly take the image of the two overlapping triangles and create two non-overlapping triangles that maintain their original properties, features, and labeling.

5.5.4 Collaboration: Overlapping Triangles (continued)

2. Explain why Dalton can or cannot use ASA to show that $\triangle RJN \cong \triangle LKN$.

Answers will vary. Although ASA is not the most direct route, he could use ASA. Using the triangle sum theorem, he knows the third angles will also be congruent, $\angle JRN \cong \angle KLN$. Once he does this, then he could use ASA.

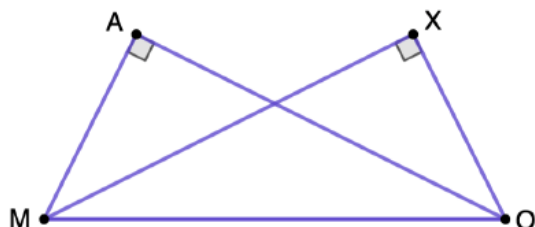
5.5.5: Try It: Drawing Conclusions From Given Information

Component Overview: Students will practice drawing conclusions from given information. For this activity, allow students to try this on their own to determine who may need additional support.



5.5.5 Try It: Drawing Conclusions from Given Information

1. Triangle MAO and triangle XOM are shown, where $\angle MAO$ and $\angle OXM$ are right angles and $\overline{MA} \cong \overline{XO}$.



Complete each statement based on the given information.

- a. I can conclude that ...

Answers will vary.

$\triangle MAO \cong \triangle OXM$

- b. I know this because ...

Answers will vary.

I can use the reflexive property of congruence to declare that $\overline{MO} \cong \overline{MO}$. Using this and the fact that these are both right triangles allows me to use the HL Theorem to prove the triangles are congruent.



Consider using this activity as a time to work with small group of students who need additional support based on their performance in the prior activities.



Have students compare their answers in question 1a and 1b with a partner. Then have each partner pair compare their work with another partner pair.

5.5.6 Wrap-Up: Ticket Out the Door

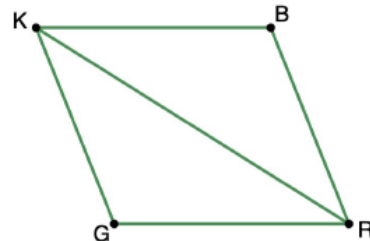
Component Overview: Students will complete a Wrap-Up that allows them to practice completing statements based on the given information. This activity is designed to be done independently to allow for an individual check for understanding on the content of this lesson.



5.5.6 Wrap-Up: Ticket Out the Door

Quadrilateral $BRGK$ is shown.

Complete each statement based on the given information.



1. Given that $\overline{KB} \cong \overline{GR}$ and $\overline{KG} \cong \overline{BR}$,
 - a. I can conclude that ...
 $\triangle KRG \cong \triangle RKB$
 - b. I know this because ...
I can use the reflexive property of congruence to know that $\overline{KR} \cong \overline{KR}$. This allows me now to use Side-Side-Side to know triangle KRG is congruent to triangle RKB.
2. Given that $\overline{KB} \cong \overline{KG}$ and $\overline{RG} \cong \overline{BR}$,
 - a. I can conclude that ...
 $\triangle KRG \cong \triangle RKB$
 - b. I know this because ...
I can use the reflexive property of congruence to know that $\overline{KR} \cong \overline{KR}$. This allows me now to use Side-Side-Side to know triangle KRG is congruent to triangle RKB.
3. Given that $\overline{KG} \parallel \overline{BR}$ and $\overline{KB} \parallel \overline{GR}$,
 - a. I can conclude that ...
 $\triangle KRG \cong \triangle RKB$
 - b. I know this because ...
When parallel lines are intersected by a transversal, alternate interior angles are congruent. Therefore, $\angle GKR \cong \angle BRK$ and $\angle GRK \cong \angle BKR$. By reflexive property, $\overline{KR} \cong \overline{KR}$ and allowing me to use Angle-Side-Angle.



The data from the student response to this activity can provide insight on whether or not the students need additional support with the content from this lesson. Questions 1 and 2 assess whether students recall the reflexive property and the conditions for SSS. Question 3 requires the student to recall alternate interior angles of parallel lines, the reflexive property, and the conditions for ASA.